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Dr. Gu

CSC112

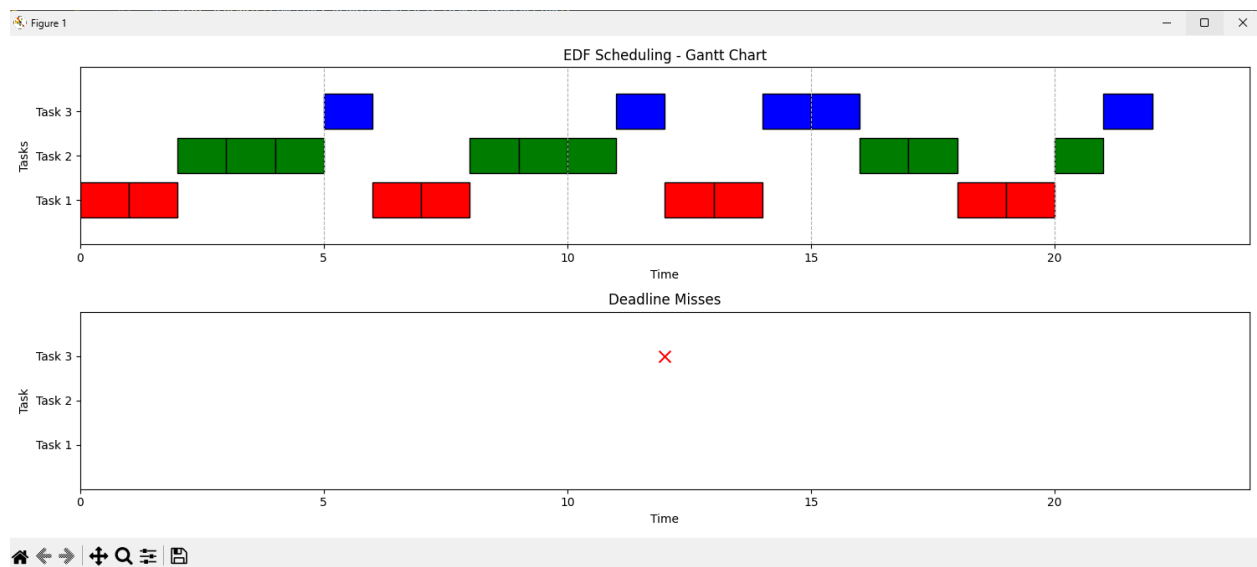
May 6, 2025

Lab 2 Reflection:

In Lab 2, we implemented two real-time scheduling algorithms, Earliest Deadline First (EDF) and Least Laxity First (LLF). These were based on the FP scheduling algorithm we were given. For scheduling under EDF, we select the job having the smallest absolute deadline in each time unit. When a job is released, we assign to it its release time plus its period as its deadline. To schedule using LLF, we calculated the laxity of each task in each time unit using the formula: $\text{Laxity} = \text{deadline} - (\text{current time} + \text{remaining execution time})$. The task having the least laxity was to run. Similar to EDF, the deadline was refreshed for each release of the jobs. We tested the two algorithms on both schedulable and unschedulable task sets.

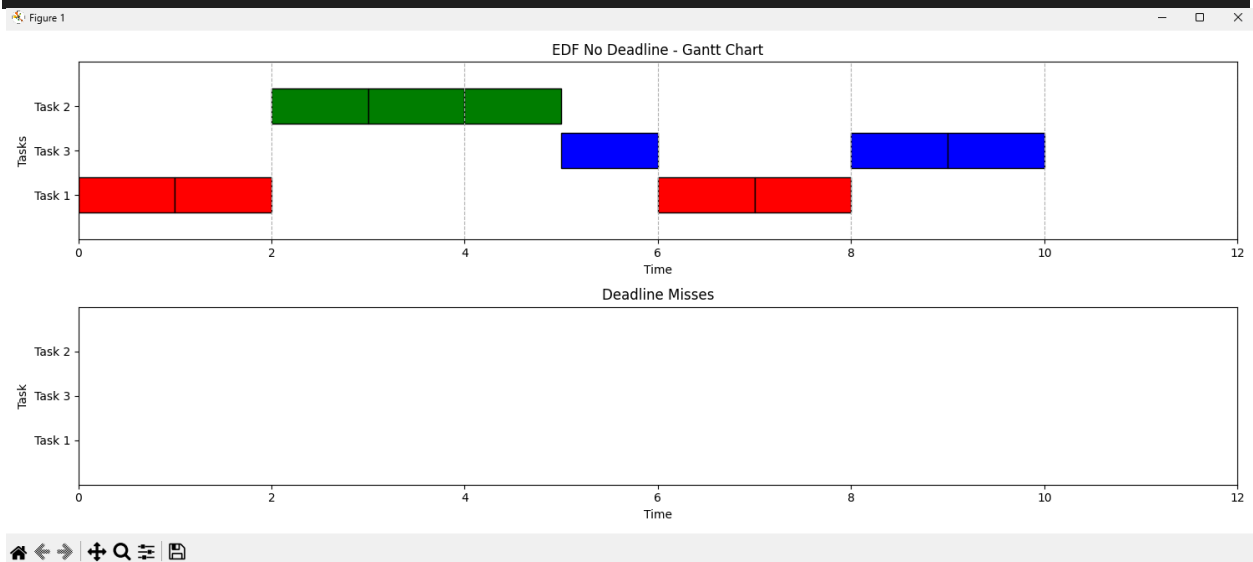
EDF Scheduling w Deadline:

```
tasks = [  
    Task(1, 6, 2, 6, 3), # Task 1: Period=6ms, Execution Time=2ms,  
    Deadline=6ms, Priority=3 (highest)  
    Task(2, 8, 3, 8, 2), # Task 2: Period=8ms, Execution Time=3ms,  
    Deadline=8ms, Priority=2  
    Task(3, 12, 3, 12, 1) # Task 3: Period=12ms, Execution Time=3ms,  
    Deadline=12ms, Priority=1 (lowest)  
]
```



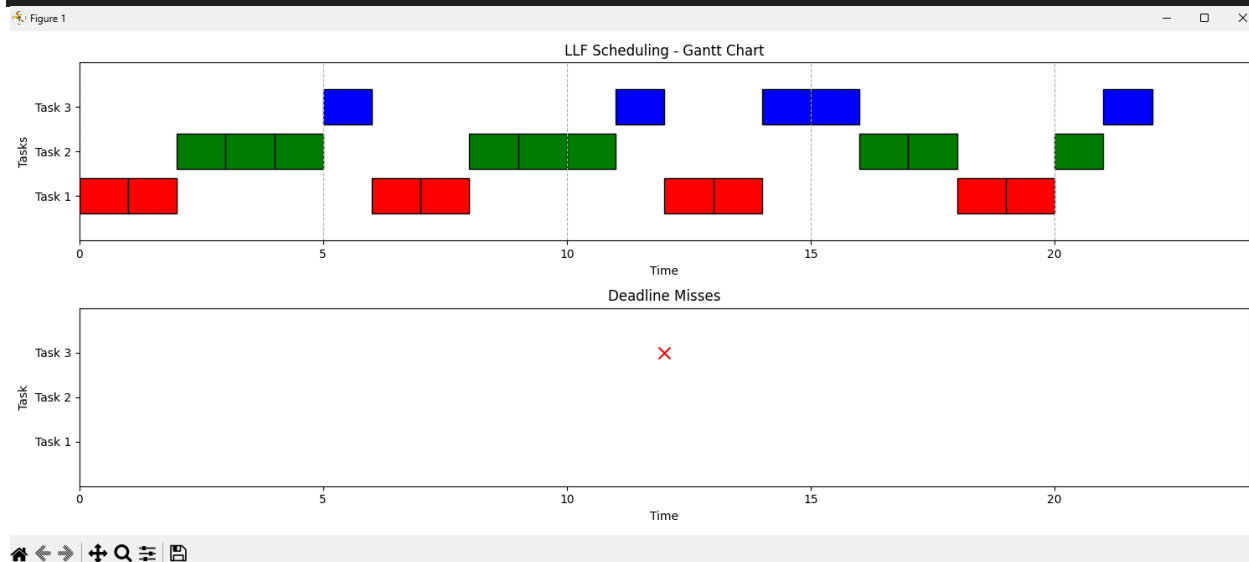
EDF Scheduling No Deadline:

```
tasks = [  
    Task(1, 6, 2, 6, 1), # Task 1: Period=6ms, Execution Time=2ms,  
    Deadline=6ms, Priority=3 (highest)  
    Task(2, 12, 3, 12, 2), # Task 2: Period=8ms, Execution Time=3ms,  
    Deadline=8ms, Priority=2  
    Task(3, 12, 3, 12, 3) # Task 3: Period=12ms, Execution Time=3ms,  
    Deadline=12ms, Priority=1 (lowest)  
]
```



LLF Scheduling with Deadline:

```
tasks = [  
    Task(1, 6, 2, 6), # Task 1: Period=6ms, Execution Time=2ms,  
Deadline=6ms, Priority=3 (highest)  
    Task(2, 8, 3, 8), # Task 2: Period=8ms, Execution Time=3ms,  
Deadline=8ms, Priority=2  
    Task(3, 12, 3, 12) # Task 3: Period=12ms, Execution Time=3ms,  
Deadline=12ms, Priority=1 (lowest)  
]
```



LLF Scheduling with No Deadline:

```
tasks = [  
    Task(1, 6, 2, 6), # Task 1: Period=6ms, Execution Time=2ms,  
Deadline=6ms, Priority=3 (highest)  
    Task(2, 12, 3, 12), # Task 2: Period=8ms, Execution Time=3ms,  
Deadline=8ms, Priority=2  
    Task(3, 12, 3, 12) # Task 3: Period=12ms, Execution Time=3ms,  
Deadline=12ms, Priority=1 (lowest)  
]
```

